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#include <stdio.h>
#include <stdlib.h>

void fcfs(int request[], int n, int head)
{
    int i, movement = 0, current = head;

    printf("\nFCFS Seek Sequence: %d ", head);

    for (i = 0; i < n; i++)
    {
        movement += abs(current - request[i]);
        current = request[i];
        printf("-> %d ", current);
    }

    printf("\nTotal Seek Time = %d\n", movement);
}

void sstf(int request[], int n, int head)
{
    int visited[100] = {0};
    int i, j, min, index;
    int movement = 0, current = head;

    printf("\nSSTF Seek Sequence: %d ", head);

    for (i = 0; i < n; i++)
    {
        min = 9999;
        index = -1;

        for (j = 0; j < n; j++)
        {
            if (!visited[j])
            {
                if (abs(current - request[j]) < min)
                {
                    min = abs(current - request[j]);
                    index = j;
                }
            }
        }

        visited[index] = 1;
        movement += min;
        current = request[index];
    }
}
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        printf("-> %d ", current);
    }

    printf("\nTotal Seek Time = %d\n", movement);
}

void scan(int request[], int n, int head, int diskSize)
{
    int left[100], right[100];
    int l = 0, r = 0;
    int i, j, temp;
    int movement = 0, current = head;

    for (i = 0; i < n; i++)
    {
        if (request[i] < head)
            left[l++] = request[i];
        else
            right[r++] = request[i];
    }

    /* Sort left */
    for (i = 0; i < l - 1; i++)
        for (j = i + 1; j < l; j++)
            if (left[i] > left[j])
            {
                temp = left[i];
                left[i] = left[j];
                left[j] = temp;
            }

    /* Sort right */
    for (i = 0; i < r - 1; i++)
        for (j = i + 1; j < r; j++)
            if (right[i] > right[j])
            {
                temp = right[i];
                right[i] = right[j];
                right[j] = temp;
            }

    printf("\nSCAN Seek Sequence: %d ", head);

    /* Move right */
    for (i = 0; i < r; i++)
    {
        movement += abs(current - right[i]);
        current = right[i];
        printf("-> %d ", current);
    }
}
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}

/* Go to end */
if (current != diskSize - 1)
{
    movement += abs(current - (diskSize - 1));
    current = diskSize - 1;
    printf("-> %d ", current);
}

/* Move left */
for (i = 1 - 1; i >= 0; i--)
{
    movement += abs(current - left[i]);
    current = left[i];
    printf("-> %d ", current);
}

printf("\nTotal Seek Time = %d\n", movement);
}

int main()
{
    int request[100];
    int n, head, diskSize;
    int i;

    printf("Enter number of requests: ");
    scanf("%d", &n);

    printf("Enter request queue:\n");

    for (i = 0; i < n; i++)
        scanf("%d", &request[i]);

    printf("Enter initial head position: ");
    scanf("%d", &head);

    printf("Enter disk size: ");
    scanf("%d", &diskSize);

    fcfs(request, n, head);
    sstf(request, n, head);
    scan(request, n, head, diskSize);

    return 0;
}
```